

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

Biology

9744/03

Paper 3 Long Structured and Free-response Questions

28 August 2018

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, Biology class and registration number on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Sections A and B

Answer **all** questions in the spaces provided on the Question Paper.

Section C

Answer any **one** question on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part of question.

For Examiner's Use	
Section A	
1	/30

This document consists of **16** printed pages.

Section A

Answer the question in this section.

- 1 (a) Chinese white poplar (*Populus tomentosa*) is a native tree species in China that grows in temperate areas. Its growth-dormancy cycle is driven by environmental cues such as temperature.

In a study in the early spring, all the mRNA from the cells in the cambium region of *P. tomentosa* were isolated on six different days in 2008 and converted to complementary DNA (cDNA). The cDNA was then used for polymerase chain reaction (PCR) analysis of the expression patterns of five cell cycle-related genes (*PtoCKS1*, *PtoCYCD3*, *PtoCYCB*, *PtoCDKB* and *PtoCDKA*) and one housekeeping gene (*UBQ-L*)

Fig. 1.1 shows the expression patterns of the six genes in the cambium region of *P. tomentosa*.

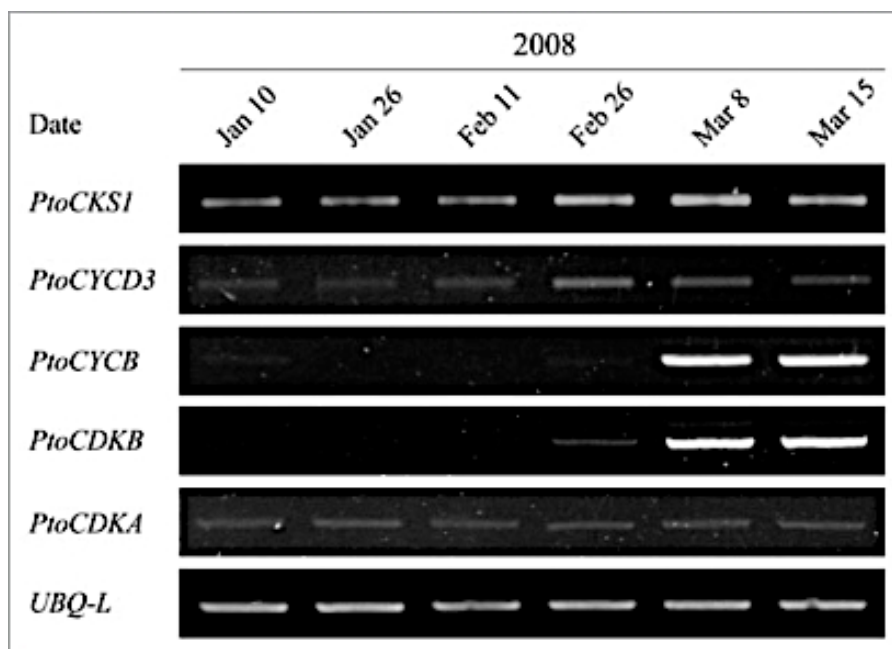


Fig. 1.1

- (i) Outline the process of PCR.

Stage 1: Denaturation

1. The reaction mix is heated to 95°C.
2. The double-stranded DNA separates into 2 single strands, as the hydrogen bonds between base pairs are broken.
3. Each strand will act as a template for the synthesis of its complementary strand.

Stage 2: Annealing

4. The reaction mix is then cooled slightly to 50 – 55°C.

5. Two types of DNA primers, the forward and reverse primers, are used.
6. Primers anneal via complementary base-pairing to the start and end of the target sequence (3' end of each separated DNA strand), marking the boundaries of the DNA to be amplified.

Stage 3: Extension/Elongation

7. The temperature is raised to 72°C.
8. Taq polymerase adds deoxyrinucleotides to the 3'-OH ends of each primer, using the DNA molecule as a template, extending the primers in the 5' → 3' direction.

[For each stage, quote temperature and one subsequent explanation for 1 mark.]

- [3]
- (ii) With reference to Fig. 1.1, describe the expression patterns of *PtoCKS1*, *PtoCYCB*, and *UBQ-L*.

1. *PtoCKS1* – Increasing expression from Jan 10 to Mar 15
2. *PtoCYCB* – No expression from Jan 10 to Feb 26, expression starts on March 8
3. *UBQ – L* – Continuous expression throughout the 8 days

[3]

- (iii) During the shift from dormancy to growth, the mitotic cell cycle occurs at an accelerated rate in the cells of *P. tomentosa*.

Outline the behaviour of chromosomes in mitosis.

1. During prophase, chromatin fibres condense into chromosomes.
2. Each duplicated chromosome appears as two identical sister chromatids joined together at their centromeres.
3. During metaphase, chromosomes line up individually at the metaphase plate.
4. During anaphase, (the centromeres divide and) the sister chromatids separate.
5. The liberated / full-fledged chromosomes begin moving toward opposite poles of the cell.
6. During telophase, chromosomes reach the poles of the cell and decondense.

[4]

(iv) Other than growth, explain **two** other functions of mitosis.

1. Mitosis maintains genetic stability of an organism.
2. This is because eukaryotic cell divisions involving mitosis produce daughter cells, each containing the same number of chromosomes and the same hereditary information as the parent cell.
3. Mitosis occurs during the repair of worn-out or damaged tissue in the body.
4. Worn-out or damaged cells are replaced by new cells that are produced by eukaryotic cell divisions involving mitosis, and are therefore genetically identical to the original cells.
5. Mitosis is the basis of asexual reproduction.
6. A single parent undergoes eukaryotic cell division(s) involving mitosis, which produces offspring genetically identical to the parent.

[2]

- (b) The increase in temperature affects leaf metabolism in several ways. It affects the activity of Rubisco, the central enzyme of photosynthesis, as well as Rubisco activase, the enzyme required for Rubisco reactivation. In addition, Rubisco activase is found to be far more sensitive to heat than Rubisco.

Fig. 1.2 shows the effect of two temperatures on the activity of the two enzymes.

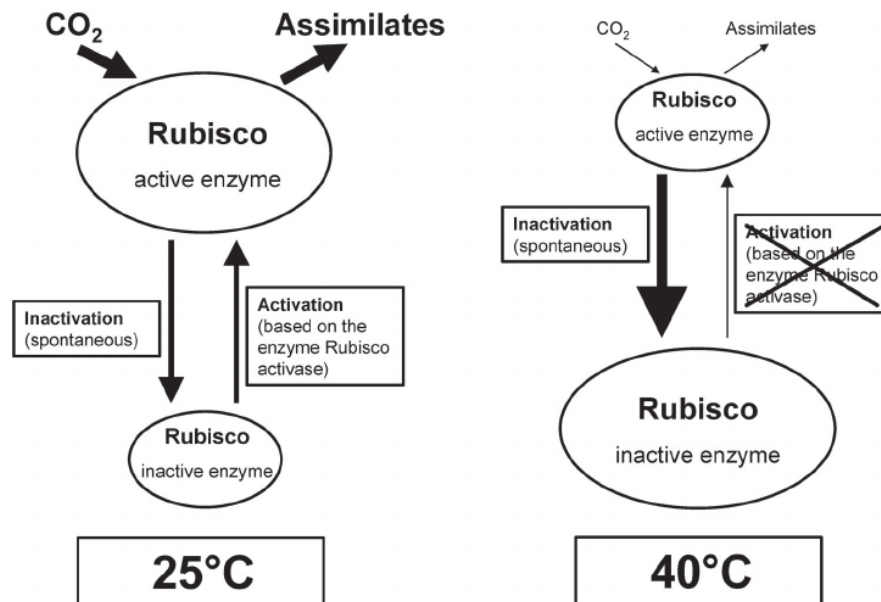


Fig. 1.2

- (i) Suggest **one** reason why Rubisco activase is far more sensitive to heat than Rubisco.

Presence of disulphide bonds in Rubisco

[1]

- (ii) Describe the effect of increasing temperature from 25°C to 40°C on Rubisco activase activity.

1. Beyond the optimum temperature, the rate of reaction decreases drastically, and eventually reaches zero.
2. The kinetic energy of the enzyme and substrate molecules continues to increase with temperature, leading to higher frequency of collisions.
3. However, the rise in temperature causes thermal agitation, disrupting weak bonds (hydrogen bonds, ionic bonds, hydrophobic interactions) holding the secondary and tertiary structures are disrupted.
4. Thus, the structure of the active sites are distorted and no longer complementary to the substrate. As a result, substrate molecules cannot bind to the active site, and ES complex cannot be formed.
5. The enzyme loses its catalytic activity progressively as enzymes are being denatured by the high temperature.

[4]

- (iii) With reference to Fig. 1.2, describe and explain the difference between the rate of photosynthesis at 25°C and that at 40°C.

1. At 25°C, there is activation of inactive rubisco to active rubisco
2. Due to functional Rubisco activase, hence rate of photosynthesis is high
3. At 40°C, there is no activation of inactive rubisco to active rubisco
4. While inactivation of active rubisco enzyme occurs spontaneously, low amount of active rubisco enzyme leads to lower rate of photosynthesis.

[3]

- (c) Other than temperature, increasing CO₂ concentration could also affect the rate of photosynthesis.

- (i) Identify and explain **one** human activity over the last few centuries that has contributed to an increase in atmospheric CO₂ concentration.

1. Burning of fossil fuel for transport/industrial/residential
2. Releases atmospheric carbon dioxide as the process combines carbon with oxygen in the air
3. Land use changes such as deforestation
4. Loss of carbon sink as carbon dioxide is taken in by trees / burning of cleared trees releases large amount of CO₂
5. Demand for meat
6. Meat production result in the release of GHGs through industrial processes and rearing of cows which increases enteric fermentation

[2]

- (ii) Explain the effects of climate change as atmospheric CO₂ concentration increases.

1. Melting of polar ice caps
2. Rising sea levels
3. Stress on fresh water supplies
4. Heat waves
5. Heavy rains
6. Death of coral reefs
7. Migration of fishes and insects,
8. Release of greenhouse gases in frozen organic matter

[Any of the above with elaboration]

[3]

- (d) Dietary deficiencies of zinc and iron are a substantial global public health problem. An estimated two billion people worldwide suffer from these deficiencies. Most of these people depend on C3 grains and legumes as their primary dietary source of zinc and iron.

A study was done to determine the concentration of zinc when C3 rice plants were grown in a controlled environment at the current atmospheric CO₂ concentration, and at the elevated atmospheric CO₂ concentration predicted for the middle of this century.

Table 1.1 shows the results of the study.

Table 1.1

concentration of zinc in C3 rice plants / mg kg ⁻¹	
grown at the current atmospheric CO ₂ concentration	grown at the elevated atmospheric CO ₂ concentration predicted for 2050
80	75
62	55
52	50

70	62
64	58
72	55
69	70
70	62

A two-sample t-test could be carried out to determine whether there is a decrease in the concentration of zinc when the C3 rice plants were grown at the elevated atmospheric CO₂ concentration predicted for 2050.

- (i) State the alternative hypothesis for the t-test.

Mean concentration of zinc in C3 plants were lower when grown at the elevated atmospheric CO₂ concentration at 2050 compared to current

[1]

- (ii) Carry out the t-test in the space provided.

Show all your working.

- 1. Workings for t-test shown, with mean values for x_1 (67.375) and x_2 (60.875)**
- 2. T-value of 1.57 / p-value of 0.0688**

[For Point 1, if GC was used, to show mean and SD values for x_1 and x_2]

[2]

(iii) State the conclusion for the t-test.

1. Do not reject the null hypothesis / no significant difference between x_1 and x_2
2. Any difference is due to chance

[2]

[Total: 30]

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Name: _____

Biology Class: 2bi2____

Section B

For Examiner's Use	
Section B	
2	/20

Section B

Answer the question in this section.

- 2 *Helicobacter pylori* is a spiral shaped, gram-negative bacterium that lives in the stomach and duodenum. In Singapore, the prevalence rate in the community (without any symptom) is estimated to increase with age — from 3% in children under the age of 5 to 71% in adults above the age of 65.

H. pylori has a genome size of approximately 1.7 Mb, with a G+C content of 35 % to 40%. Its genome sequence is highly variable. The genome of strain 26695 includes 1,587 genes, whereas the genome of strain J99 includes only 1,491 genes. Many strains carry one or more plasmids, which do not seem to carry antibiotic resistance genes or virulence genes.

- (a) Compare the structure and organisation of *H. pylori* genome with its host cell genome.

Similarities:

- S1.ref to double-stranded nature of both genome;**
S2.ref to DNA nature of both genome;

Differences:

- D1.*H.pylori* genome is packed in circular chromosome while host genome is packed into multiple linear chromosomes;**
D2.*H.pylori* DNA is organised into supercoiled loops associated with nucleoid proteins, while host DNA is wound around histones and arranged to form coiled coils in

[4]

Fig. 2.1 shows the major virulence and colonization factors of *H. pylori*. *H. pylori* uses its flagella and chemotaxis mechanisms to swim through the mucus lining the gastric epithelium. It attaches to the host cell surface and deliver toxins, such as CagA and VacA, into the host cell to control various aspects of host cell function.

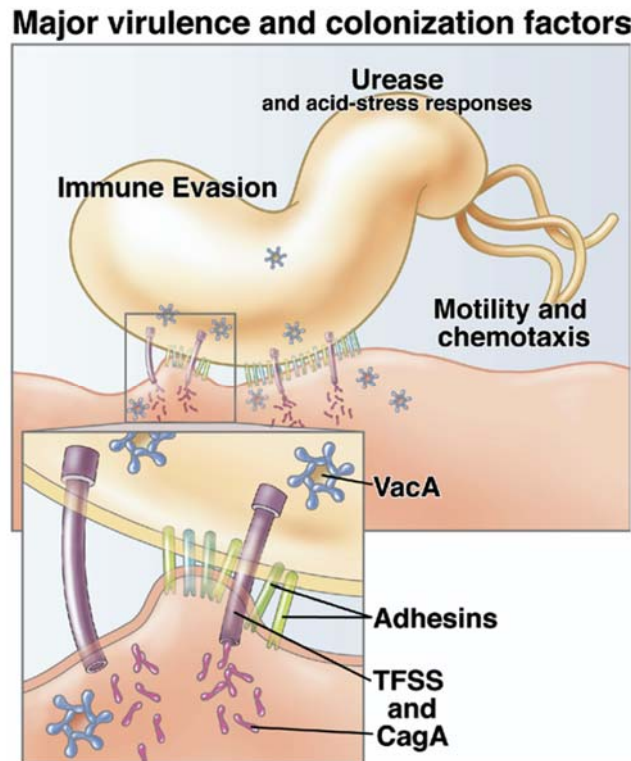


Fig. 2.1

- (b) There are several ways to diagnose *H. pylori*. One method is to draw a sample of blood from the patient and test the blood for presence of *H. pylori*-specific antibody.

Explain how the presence *H. pylori* triggers the production of *H. pylori*-specific antibodies in an infected patient.

1. *H. pylori* engulfed by dendritic cells/macrophages at the gastric epithelium and down by lysosomes to produce antigenic peptides;
2. Dendritic cell presents antigenic peptides to helper T cells and activates helper T cells;
3. Naive B cells internalizes and presents antigenic peptides;
4. TCR on helper T cell recognises and binds antigen on B cell and stimulates B cell to proliferate;
5. Some B cells differentiate into plasma cells to produce *H. pylori*-specific antibodies;

[3]

The first line therapy for *H. pylori* infection is known as triple therapy, comprising two antibiotics, clarithromycin and amoxicillin, and one proton pump inhibitor. All three drugs are administered two times a day for 14 days. Amoxicillin is similar in action to penicillin while clarithromycin binds to 23S rRNA.

(c) Discuss the deliberate use of the two antibiotics in the therapy.

1. Amoxicillin inhibits cell wall synthesis by inhibiting the formation of cross-links between adjacent chains of peptidoglycan; [4]
2. Use of amoxicillin will result in lysis of newly divided bacterial cells
3. Clarithromycin prevents formation of functional ribosomes;
4. Use of clarithromycin prevents bacterial cells from producing necessary proteins, such as CagA, thus reducing survival of existing bacterial cells;
5. Ref to high mutation rate of bacteria + develop resistance to one of the antibiotics during course of treatment;
6. Ref to presence of second antibiotic allows resistant bacteria to be killed

Early studies found that patients with antibodies against CagA showed higher rates of gastric adenocarcinoma. The C-terminal tail of the CagA is a target for phosphorylation by kinases coded by cellular oncogenes. After phosphorylation, CagA would trigger signals that resemble the activation of receptor tyrosine kinase by growth factors, controlling proliferative activities, adhesion, and cytoskeletal organization of epithelial cells. Other studies have shown that CagA leads to increased degradation of p53.

(d) Using the information provided, explain why patients with prolonged *H. pylori* infection have an increased risk of developing adenocarcinoma.

1. Increased degradation of p53 leads to decreased repair of damaged DNA;
2. Leads to accumulation of mutations;
3. Higher chance for proto-oncogenes to accumulate gain-of-function mutation to become oncogene;
4. Oncogenes phosphorylate CagA leading to greater cell proliferation even in absence of growth factor;
5. Resulting cytoskeletal organisation may lead cells to demonstrate loss of contact of inhibition thus forming tumor

[4]

cagA is a gene that has many variants. In particular, there are different numbers of repeat sequences located in the 3' region of the *cagA* gene of different *H. pylori* strains. Each repeat region of the CagA protein contains Glu-Pro-Ile-Tyr-Ala (EPIYA) motifs, including a tyrosine phosphorylation site. Studies confirmed that, in western countries, the incidence of gastric cancer is notably higher in patients infected with strains containing multiple EPIYA-C segments than in patients infected with strains containing a single EPIYA-C segment.

Fig. 2.2 shows three variants of CagA protein.

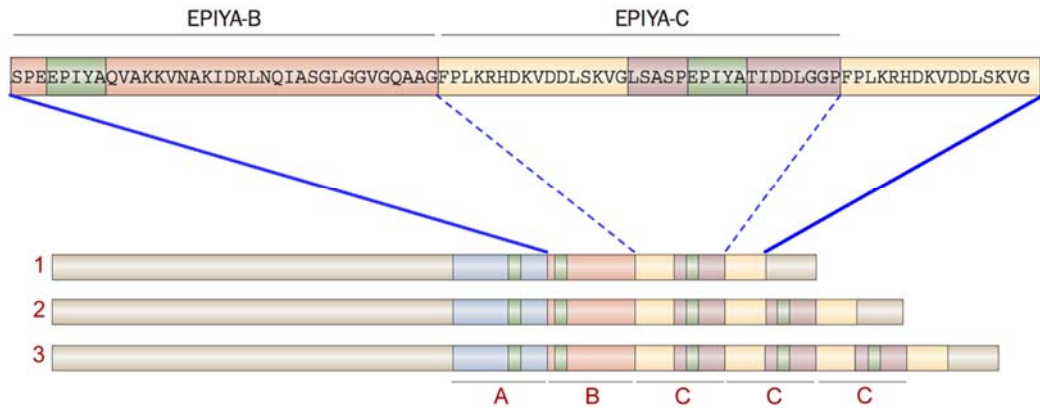
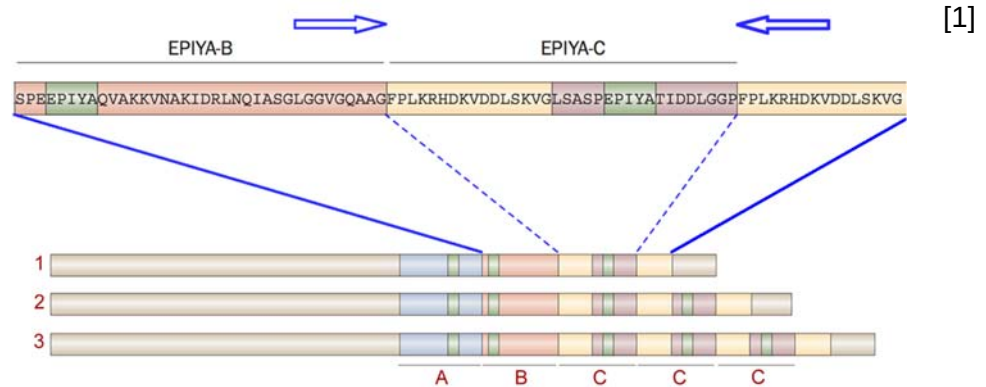


Fig. 2.2

- (e) (i) Polymerase chain reaction (PCR) could be carried out on a tissue sample to determine the *cagA* variant of the *H. pylori* strain infecting the patient.

Indicate on Fig. 2.2, the ideal position of **one** set of primers to distinguish the DNA sequence corresponding to the three CagA variants by PCR.



- (ii) Determine the size difference between PCR products from tissue samples containing variant 1 and variant 2 respectively.

Number of amino acid difference = 34

Number of nucleotide difference = 34 x 3 = 102

[1]

- (iii) Outline the principle and procedure of a molecular technique that could be used to determine the variant of *cagA* gene after PCR has been performed.

1. Agarose gel electrophoresis + separation of nucleic acid by size;
2. ref to migration of DNA molecules towards positive electrode/anode;
3. ref to molecular ladder + mixture of DNA fragments of known sizes added to same gel to estimate size of PCR fragment
4. ref to setting up gel electrophoresis

[3]

[Total: 20]

Name: _____

Biology Class: 2bi2____

Section C

For Examiner's Use	
Section C	
___ (a)	/15
___ (b)	/10

Section C

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated with large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 3** **(a)** Using **three** named examples, explain the relationship between the structures and functions of carbohydrates in living organisms. [15]
- (b)** Describe how photosynthesis and cellular respiration complement each other in the environment, and compare the two processes. [10]

[Total: 25]

- 4** **(a)** Using **three** named examples, explain the relationship between the structures and functions of proteins in living organisms. [15]
- (b)** Describe how enzymes increase the rate of chemical reactions, and compare the actions of competitive and non-competitive enzyme inhibitors. [10]

[Total: 25]

Mark Scheme for FRQ 3(a):

<i>Starch / Glycogen</i>	<i>1 mark for each valid example: Starch / Glycogen</i>	2
	<i>Any 7 of the following:</i> 1. Glycosidic bonds (are strong covalent bonds that) join many α-glucose monomers together to form starch / glycogen; 2. Which can be released upon hydrolysis as respiratory substrates; 3. This also forms a large molecule so that the molecule is insoluble in water; 4. α-1,4-glycosidic bonds forms a helical structure; 5. which causes the molecule to be compact for storage; 6. Hydroxyl groups of glucose residues project into the interior of the helices; 7. Hence, absence of hydrogen bonds with water causing starch / glycogen to be insoluble in water; 8. Hence, they can be stored in large quantities without affecting the osmotic potential of cells; 9. α-1,6-glycosidic bonds allows for amylopectin / glycogen to be highly branched; 10. Hence, a greater amount of carbohydrates can be stored per unit volume; 11. It also allows for many enzymes to act on it at the same time; 12. Allows for quick release of glucose (for an increased rate of respiration);	7
<i>Cellulose</i>	<i>1 mark for valid example: Cellulose</i>	1
	<i>Any 4 of the following:</i> 13. Glycosidic bonds (are strong covalent bonds) join many β-glucose monomers together to form cellulose; 14. This also forms a large molecule so that the molecule is insoluble in water; 15. β-1,4-glycosidic bonds allow the formation of straight chains; 16. Hydrogen bond cross links between hydroxyl groups of adjacent chains; 17. prevent the hydroxyl groups from forming hydrogen bonds with water hence allowing it to be insoluble in water; 18. Also allows for formation of microfibrils and macrofibrils; 19. which allows cellulose to have tremendous tensile strength; 20. Allowing it to perform its function as a structural molecule; 21. to help prevent cells from bursting / maintain shape of cell / allows for cell turgidity;	4
<i>QWC</i>	<i>Attempt to relate structure to function for at least two named examples</i>	1

Mark Scheme for FRQ 3(b):

Complement	Any 2 of the following:				2
	1. Photosynthesis makes the glucose that is used in cellular respiration to make ATP, while the glucose in cellular respiration is then turned back into carbon dioxide, which is used in photosynthesis; 2. Water is broken down to form oxygen during photosynthesis, while oxygen is combined with hydrogen to form water in cellular respiration; 3. Photosynthesis requires carbon dioxide and releases oxygen, while cellular respiration requires oxygen and releases carbon dioxide;				
Similarities	Any 2 valid similarities:				2
	4. ATP synthase catalyses the formation of ATP during chemiosmosis; 5. Electron transport chain is required for generation of an electrochemical proton gradient; 6. AVP				
Differences	Any 5 valid differences (different from points in "Complement" section):				5
			photosynthesis	cellular respiration	
	7.	cellular location	chloroplast	cytoplasm (anaerobic) & mitochondria (aerobic)	
	8.	equation	$6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$	$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP (energy)}$	
	9.	stages	2 (light-dependent and light-independent reactions)	4 (glycolysis, link reaction, Krebs cycle, oxidative phosphorylation)	
	10.	establishment of proton gradient	protons pumped from stroma across thylakoid membrane into thylakoid space	protons pumped from mitochondrial matrix across inner mitochondrial membrane into intermembrane space	
	11.	main function	production of food / energy capture / anabolic	breakdown of food / energy release / catabolic	
	12.	AVP			
QWC	Use of table to compare photosynthesis and cellular respiration				1

Mark Scheme for FRQ 4(a):

<i>Haemoglobin</i>	1 mark for valid example: Haemoglobin (transport)	1
	<i>Any 4 of the following:</i> 1. Amino acids are joined together by (strong covalent bonds known as) peptide bonds (to form the alpha and beta chains which are 141 amino acids and 146 amino acids long respectively); 2. Hydrogen bonds are formed between the –CO and –NH groups of the polypeptide backbone to form α-helices; 3. The 3D conformation of the protein is maintained by ionic, hydrogen bonds and hydrophobic interactions between R-groups (at least 2 bonds); 4. The two $\alpha\beta$ dimers are held together mainly by weak hydrogen bonds and ionic bonds; 5. In contrast, the two polypeptide chains within each $\alpha\beta$ dimer are tightly held together by mainly hydrophobic interactions; 6. This allows for the arrangement of hydrophobic amino acid residues within the interior of the globular structure; 7. allowing for the formation of a haem binding pocket/ hydrophobic environment / deep hydrophobic cleft for the haem group to bind to oxygen; 8. Hydrophilic amino acid residues are found at the surface of the globin; 9. Allowing hydrogen bonds to be formed between the water and hydrophilic amino acid residues; 10. Allows for solubility in a aqueous medium/ allows it to be a good transporter of oxygen in blood; 11. The hydrogen bonds between the two $\alpha\beta$ dimers allow for cooperativity to occur; 12. When an oxygen molecule binds to / is released from one haemoglobin subunit, the binding / release induces a conformational change in the remaining subunit; 13. Which increases / lowers the affinity for oxygen of the remaining three oxygen binding sites respectively; 14. Hence facilitates the loading and unloading of oxygen;	4

<i>Collagen</i>	1 mark for valid example: Collagen (structural)	1
	<i>Any 4 of the following:</i> 15. Amino acids join together by (strong covalent bonds known as) peptide bonds (to form the alpha chain); 16. which has a repeating tri-peptide sequence of glycine-X-Y; 17. where X is often proline and Y is often hydroxyproline or hydroxylysine; 18. Polymerisation allows for the formation of a long molecule that allows collagen to function as a structural protein; 19. Collagen is insoluble in water due to absence of hydrogen bonds with water; 20. As glycine and proline are hydrophobic; 21. Collagen has high tensile strength; 22. Due to covalent cross-links between different tropocollagen molecules; 23. Cross linking by hydrogen bonds between alpha chains / within triple helix / tropocollagen (Reject: within collagen); (award mark if linked to insolubility and tensile strength); 24. Bundling for the formation of a fibril and subsequently fibres;	4
<i>G-protein linked receptor</i>	1 mark for valid example: G-protein linked receptor (signalling)	1
	<i>Any 3 of the following:</i> 25. Amino acids join together by (strong covalent bonds known as) peptide bonds (to form the alpha chain); 26. The presence of hydrogen bonds formed between the –CO and –NH groups of the polypeptide backbone to form 7 α-helices; 27. The 3D conformation of the protein is maintained by ionic, disulfide, hydrogen bonds and hydrophobic interactions between R groups (at least 2 bonds); 28. This allows for the arrangement of hydrophobic / non-polar R groups of amino acids facing exterior of α helices to interact with hydrophobic / hydrocarbon tails of the phospholipid bilayer; 29. hence allowing for embedment of the protein in the membrane; 30. This also allows for the formation of a binding site to allow binding of a signal molecule/ ligand; 31. And the formation of a binding site to allow binding of the G protein; 32. Hence allowing for the signal molecule to bind and trigger conformation change in GPLR; 33. And activated GPLR can bind to the G protein and activate G protein when GTP displaces GDP;	3

QWC	Attempt to relate structure to function for at least two named examples	1
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Mark Scheme for FRQ 4(b):

How enzymes increase the rate of reactions	Any 2 of the following: 1. By lowering of the energy barrier and hence activation energy of the reaction; 2. Binding of substrates to enzyme active site brings substrates into close proximity to each other; 3. In enzyme active site, substrates undergo slight distortion and this strains bonds in the substrate that need to be broken; 4. Substrates are held in the correct orientation such that the bonds in the substrate are exposed to chemical attack; 5. Enzymes create the appropriate microenvironment for a reaction to occur; 6. e.g. hydrophobic amino acids in enzyme create a water-free zone that allows non-polar reactants to react more easily / AVP; 7. R groups of acidic and basic amino acids in enzyme active site facilitate reaction between substrates; 8. to increase chances / rate of a reaction; (note: mark given if one of the points from point 2 to 7 are mentioned)	3
Similarity	Any 1 valid similarity: 9. When the inhibitor binds to the enzyme's active site or site away from active site, an enzyme-inhibitor (E-I) complex is formed; 10. The rate of the enzyme-catalysed reaction is reduced; 11. AVP	1
Differences	Any 5 valid differences (on next page)	5
QWC	Use of table to compare the actions of competitive and non-competitive enzyme inhibitors	1

Competitive Inhibition	Non-competitive Inhibition
○ A competitive inhibitor, as its name suggests, competes with the genuine substrate for binding to the enzyme's active site. ○ To do so, it must have a structural resemblance to the genuine substrate.	○ Unlike a competitive inhibitor, a non-competitive does not compete with the genuine substrate for binding to the enzyme's active site. ○ This is because it has no structural resemblance to the genuine substrate.
○ Binds loosely to the enzyme's active site	○ Binds loosely to the enzyme away from the active site
○ Binding of the competitive inhibitor to the enzyme's active site prevents binding of the genuine substrate to the enzyme's active site. ○ This decreases the number of active sites available for the genuine substrates to bind and form enzyme-substrate complexes, and eventually, products.	○ Binding of the non-competitive inhibitor to the site away from the active site does not prevent binding of the genuine substrate to the enzyme's active site, and vice versa. ○ When both the genuine substrate and the non-competitive inhibitor are bound, the enzyme-substrate-inhibitor (ESI) complex is formed. Products cannot be formed as the ESI complex can only be converted back to the ES complex or the EI complex. This simply prevents product formation for a limited time.
○ Increasing substrate concentration may reduce or even completely remove the effect of the competitive inhibitor. ○ With more substrate molecules present, there is a higher probability of them displacing the weakly associated inhibitor molecules from the active sites and forming enzyme-substrate complexes. (substrate will "out-compete" inhibitor) ○ With high enough substrate concentration, $V_{\max}$ of uninhibited reaction can be reached.	○ Increasing substrate concentration may reduce but cannot completely remove the effect of the non-competitive inhibitor. ○ No matter how high the concentration of substrate is, some of the enzymes will still be inhibited as the non-competitive inhibitor does not bind to the same site as the genuine substrate. ○ Even with very high substrate concentration, $V_{\max}$ of the uninhibited reaction can never be reached. ○ Rate of reaction can only be increased with an increase in enzyme concentration while keeping inhibitor concentration constant.
○ $V_{\max}$ of inhibited reaction is the same as that of uninhibited reaction. ○ K_m for inhibited reaction is higher than that for uninhibited reaction.	○ $V_{\max}$ of inhibited reaction is lower than that of uninhibited reaction. ○ K_m for inhibited reaction is the same as that for uninhibited reaction.